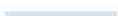


Validation, Cross-Validation and Tuning

Make model decisions with valid boundaries
and decision-aware metrics.



Learning outcomes

01

Choose splits that reflect future use.

02

Tune without contaminating the test set.

03

Search hyperparameters within a reproducible budget.

The concept system

01

holdout

02

cross-validation

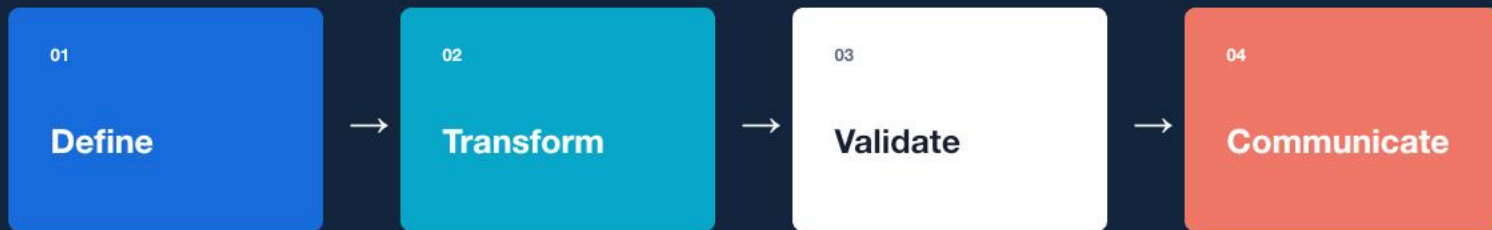
03

time splits

04

hyperparameters

A repeatable reasoning loop



Design validation schemes for random and temporal datasets.

Guided lab

LAB BRIEF

Design validation schemes for random and temporal datasets.

[workbook](#) / [tests](#) / [evidence](#)

01

Frame

Define inputs, outputs, and success criteria.

02

Build

Implement the smallest correct workflow.

03

Verify

Test normal, boundary, and failure cases.

04

Explain

Record the evidence and remaining limits.

Portfolio deliverable

A validation protocol and reproducible tuning run.

QUALITY GATE

- **Reproducible**
- **Tested**
- **Decision-relevant**

WEEKLY ASSESSMENT

Evaluation Decision Memo

Validity 35%, metric reasoning 25%, error analysis 20%, uncertainty 10%, communication 10%.

What should remain after the lesson?

CONCEPT

holdout + cross-validation

PRACTICE

Design validation schemes for random and temporal datasets.

EVIDENCE

A validation protocol and reproducible tuning run.
